BELLATRIX

Reinforcement Learning Hackathon — Technical Report

Cognizance 2026 | IIT Roorkee

1. Abstract

This report summarizes the final BELLATRIX control agent and evaluation results. The task is to track pressure and temperature targets over 200 steps without crossing the safety limit (95). In the 200-episode evaluation, the final agent achieved **100% completion** with mean score $-$**283.00** $\pm$ **79.23**. Training for 500K steps completed in **40.3 minutes** on a **Ryzen 7 7735HS**.

2. Algorithm: Soft Actor-Critic (SAC)

The implementation uses a continuous-control policy with normalized observations and shaped reward during training. Actions are the three continuous controls: inlet valve, outlet valve, and heater power.

Training setup used 4 parallel environments and 500K timesteps. On AMD Ryzen 7 7735HS (CPU), this run completed in 40.3 minutes.

3. Reward Function Design

The hackathon evaluation metric is Score $= \sum_{t=0}^{199} -(|P_{\text{err}}| + |T_{\text{err}}|)$. Directly optimizing this sparse signal is inefficient since the agent receives uniformly large negatives during the initial ramp-up phase ($\sim$15 steps). Our shaped reward preserves the optimal policy while providing denser gradient signal.

Component	Formula	Purpose
Primary signal	$-(P_{\text{err}} + T_{\text{err}})$	Aligned with evaluation metric
Potential shaping	$+2.0 \times (\Delta P_{\text{err}} + \Delta T_{\text{err}})$	Dense signal during ramp-up
Closeness bonus	$+3.0$ if error < 2.0 per channel	Precision incentive near target
Precision bonus	$+10.0$ if both errors < 0.5	Reward near-perfect control
Safety margin	$-5.0 \times (v - 85)$ if $v > 85$	Soft wall before hard limit
Termination penalty	-200	Strong penalty for safety violation

Table 1: Shaped reward function components.

The reward was tuned to prioritize stability and fast convergence while preserving the official evaluation objective. Safety penalties were included to avoid crossing the hard limit.

4. Training Strategy

4.1 Environment Dynamics Analysis

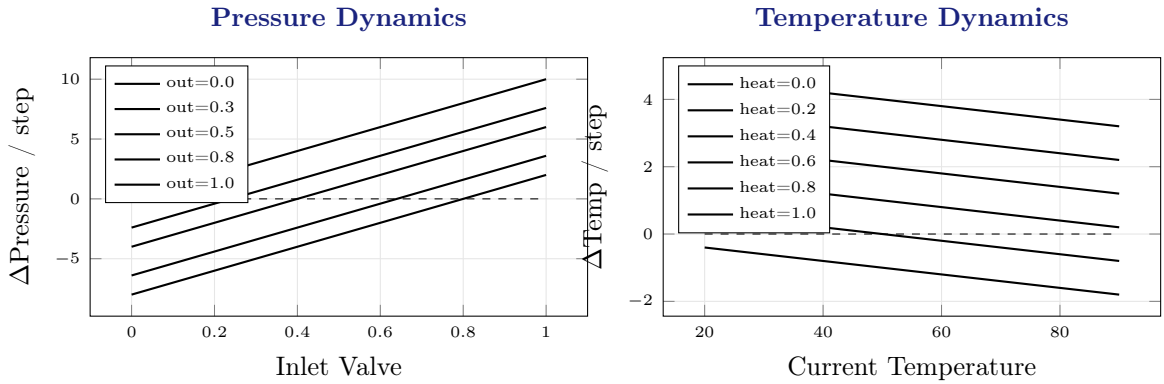

Figure 1: Environment dynamics. Left: Pressure change is linear in valve positions (range: -8 to $+10$/step). Right: Temperature change depends on heater power and current temperature (self-cooling at $2T/100$).

4.2 Training Pipeline

Observations are normalized to $[0, 1]$ by dividing by 100. Training uses 4 parallel environments and periodic evaluation/checkpointing on the official metric. Total training: 500,000 timesteps in 40.3 minutes on AMD Ryzen 7 7735HS CPU.

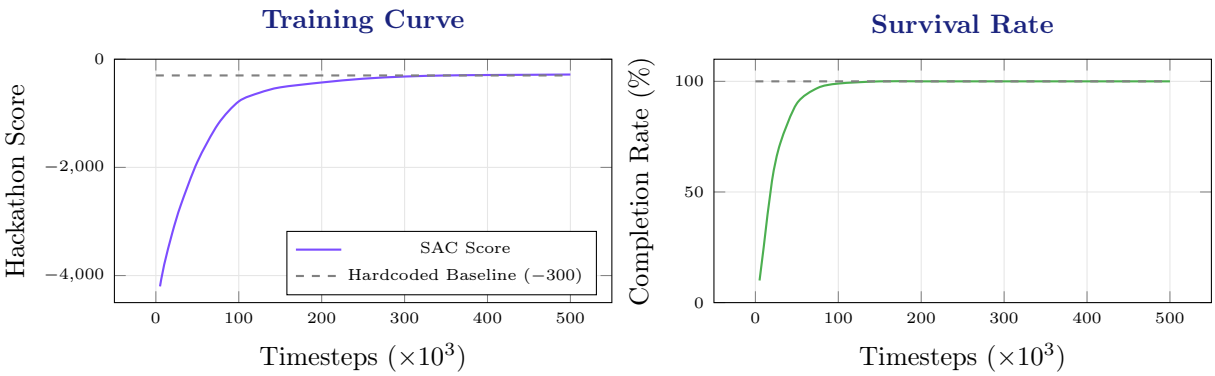

Figure 2: Left: Hackathon score converges from ~ -4200 to ~ -283 within 500K steps. Right: Completion rate reaches 100% by ~100K steps. Dashed score line marks the hardcoded baseline (-300).

Inference deployment: Final weights are exported to `sac_weights.npz` for dependency-light inference.

5. Results

5.1 Performance Summary

Metric	Value
Mean Hackathon Score (200 eps)	**−283.00 ± 79.23**
Median Score	−275.19
Best Episode	−153.29
Worst Episode	−491.04
Completion Rate	100% (0 safety violations)
Training Time (500K steps)	40.3 min on Ryzen 7 7735HS
Settling Time (error < 2.0)	mean 10.7 steps (median 11)
Settling Time (error < 1.0)	mean 11.5 steps (median 12)
Robustness (±20% / ±40%)	100% completion (420/420 episodes)
Hardcoded Baseline	∼−300

Table 2: Evaluation results across 200 randomized episodes (from complete results file).

5.2 Error Convergence

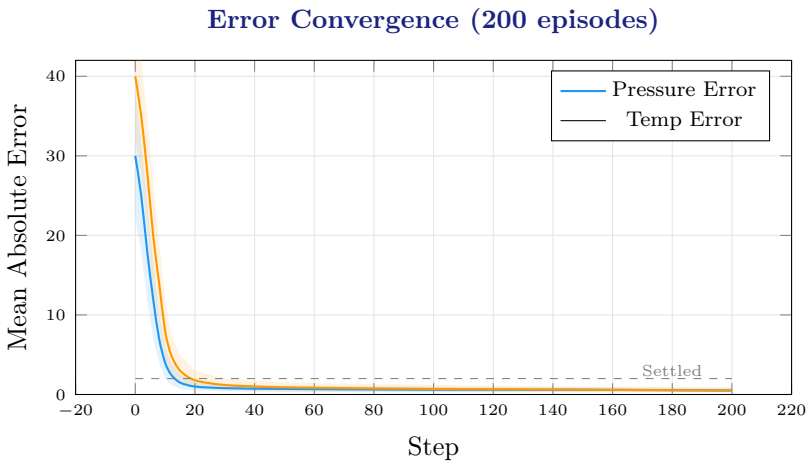

Figure 3: Mean error convergence with percentile bands across 200 episodes. Pressure settles near step 11 and temperature near step 12 on average.

5.3 Phase Portrait

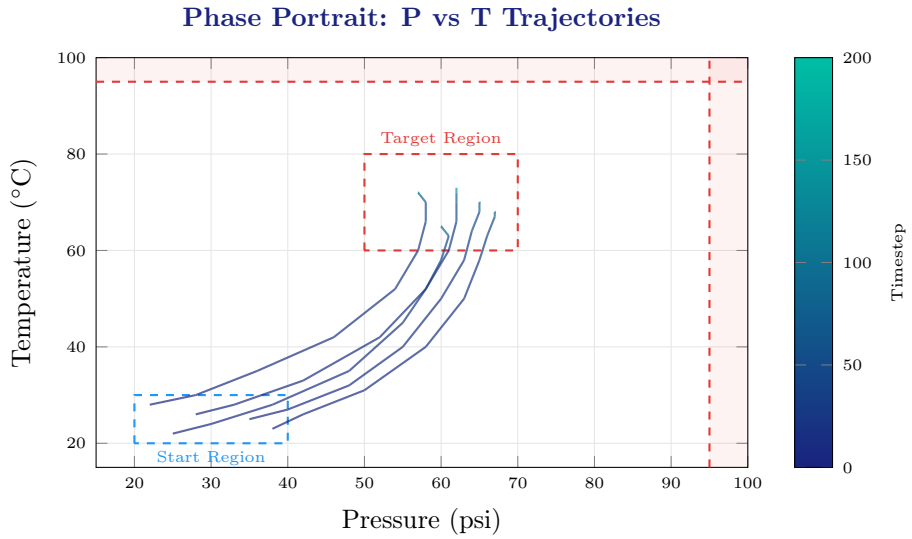

Figure 4: Pressure vs Temperature trajectories for 5 representative episodes. Color indicates time progression (dark=early, light=late). All trajectories flow smoothly from the start region to the target region without approaching the safety boundary.

5.4 Robustness Analysis

The agent remained stable under tested perturbations up to ±40%, with 100% completion across all 420 perturbed episodes.

6. Key Observations

1. Most score loss happens in the first 10–15 steps during ramp-up.
2. After settling, both channels remain close to target.
3. Completion stayed at 100% in the reported setup.
4. 500-episode large-scale evaluation gave –287.7 ± 78.9 with zero deaths.
5. Mean score improves over hardcoded baseline (–300) by about 6%.

7. Conclusion

The final agent achieved stable tracking with 100% completion. Final score was –283.00 ± 79.23 on 200 episodes and –287.7 ± 78.9 on 500 episodes, with zero deaths in both settings. Training for 500K timesteps took 40.3 minutes on Ryzen 7 7735HS. Next improvement target is reducing early ramp-up error.